Name: _____

WORKSHEET

Decimal representation

1 Convert the following terminating decimals to fractions, expressing your answer in simplest form.

a 0.344

b 0.05025

c 1.2736

d 32.43715

e 0.860052

f 0.19864

2 Convert the following fractions to terminating decimals.

a $\frac{21}{24}$

b $\frac{39}{80}$

c $\frac{12}{125}$

d $\frac{25}{16}$

e $\frac{17}{2500}$

f $\frac{171891}{50000}$

3 Convert the following fractions to recurring decimals.

a $\frac{4}{15}$

b $\frac{41}{36}$

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c $\frac{9}{110}$

e $\frac{67}{54}$

d $\frac{125}{999}$

f $\frac{12}{65}$

4 Copy and complete this conversion of $0.\overline{185}$ to a fraction.

$x = 0.185185185185185....$

$1000x =$ _____

$-x = 0.185185185185185.....$

$999x =$ _____

$x = \frac{\quad}{999}$

$=$

5 Convert the following recurring decimals to fractions.

a 0.4545454545.....

d 0.0833333333.....

b 0.703703703703.....

e 0.98148148148.....

c 0.24444444.....

f 0.857142857142857....

6 Convert the following fractions to decimals.

a $\frac{19}{26}$

b $\frac{156}{125}$

c $\frac{13}{16}$

d $\frac{29}{35}$

e $\frac{683}{750}$

f $\frac{5}{44}$

7 Convert the following decimals to fractions.

a 0.8525

b $0.\overline{69}$

c $0.\ 28125$

d $0.\overline{592}$

e $0.\overline{428571}$

f $0.03\overline{571428}$

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Answers

1 **a** $\frac{43}{125}$

b $\frac{201}{4000}$

c $\frac{796}{625}$

d $\frac{648743}{20000}$

e $\frac{215013}{250000}$

f $\frac{2483}{12500}$

2 **a** 0.875

b 0.4875

c 0.96

d 1.5625

e 0.0068

f 3.43782

3 **a** $0.2\dot{6}$

b $1.13\dot{8}$

c $0.0\overline{81}$

d $0.\overline{125}$

e $1.2\overline{407}$

f $0.1\overline{846153}$

4
$$x = 0.185185185185185....$$
$$1000x = 185185185185...$$
$$-x = 0.185185185185185....$$
$$999x = 185$$
$$x = \frac{185}{999}$$
$$= \frac{5}{27}$$

5 **a** $\frac{5}{11}$

b $\frac{19}{27}$

c $\frac{11}{45}$

d $\frac{1}{12}$

e $\frac{53}{54}$

f $\frac{6}{7}$

6 **a** $0.7\overline{307692}$

b 1.248

c 0.8125

d $0.8\overline{285714}$

e $0.91\dot{0}6$

f $0.11\overline{36}$

7 **a** $\frac{341}{400}$

b $\frac{23}{33}$

c $\frac{9}{32}$

d $\frac{16}{27}$

e $\frac{4}{7}$

f $\frac{1}{28}$

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